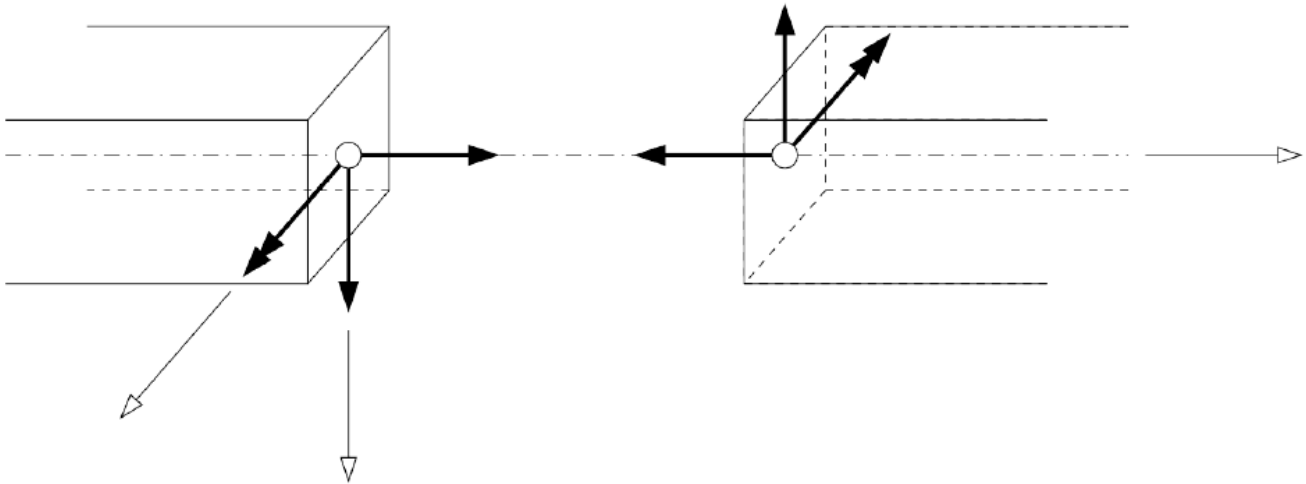
Engineering Mechanics I

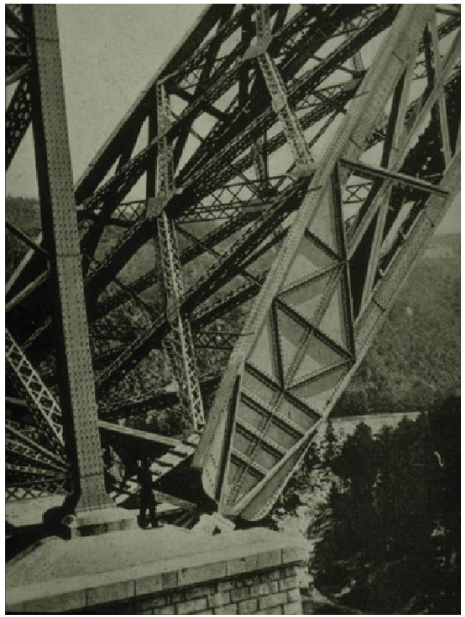
Lecture

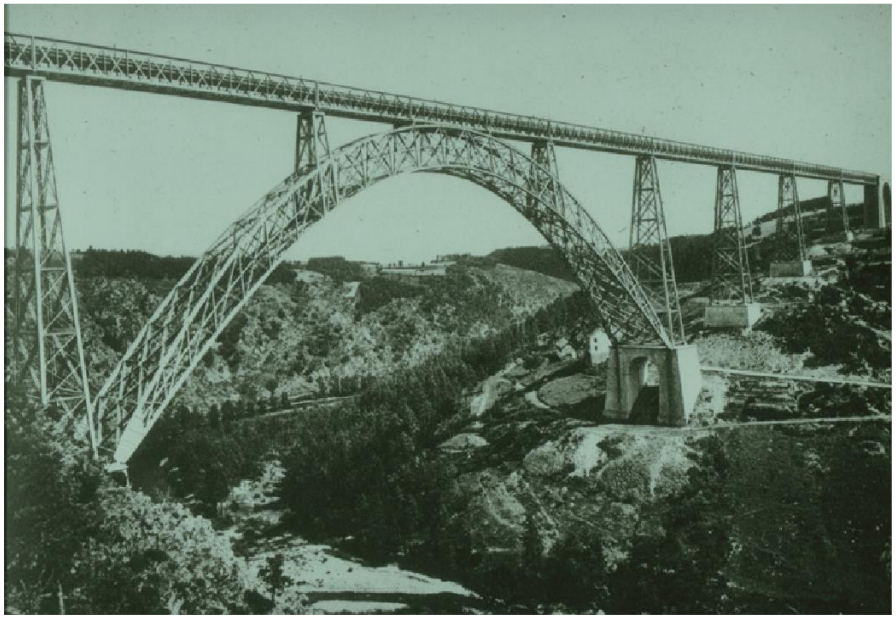


6. Trusses

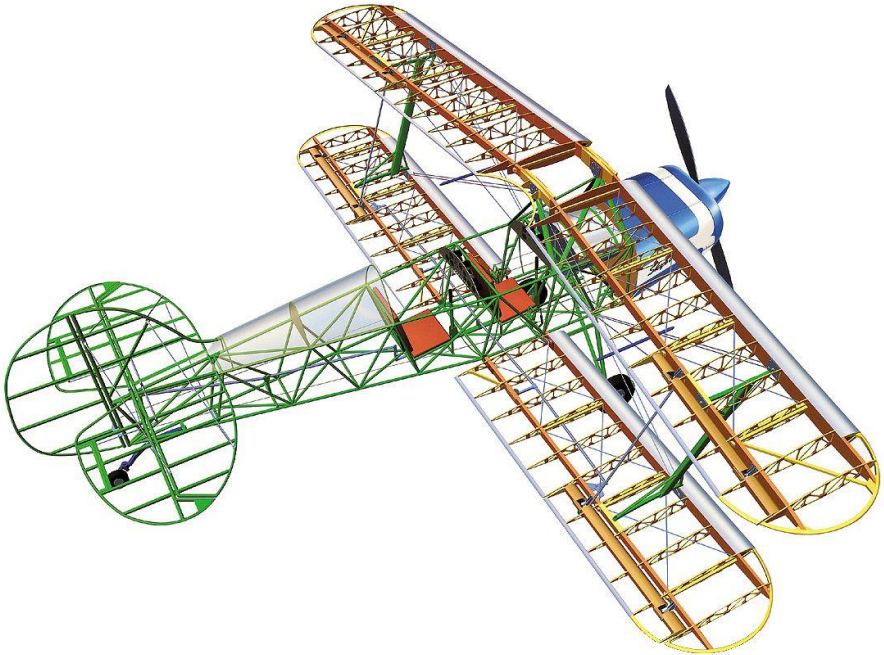


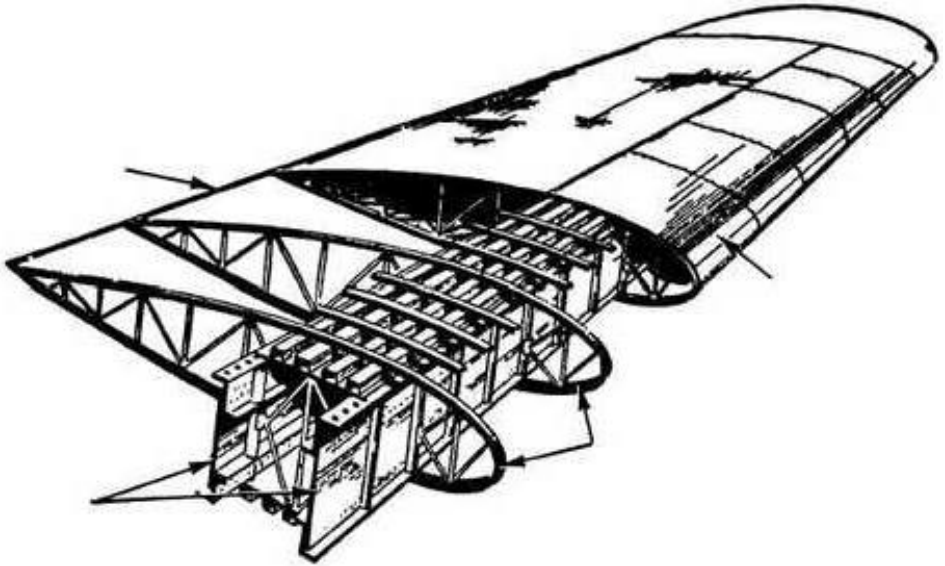




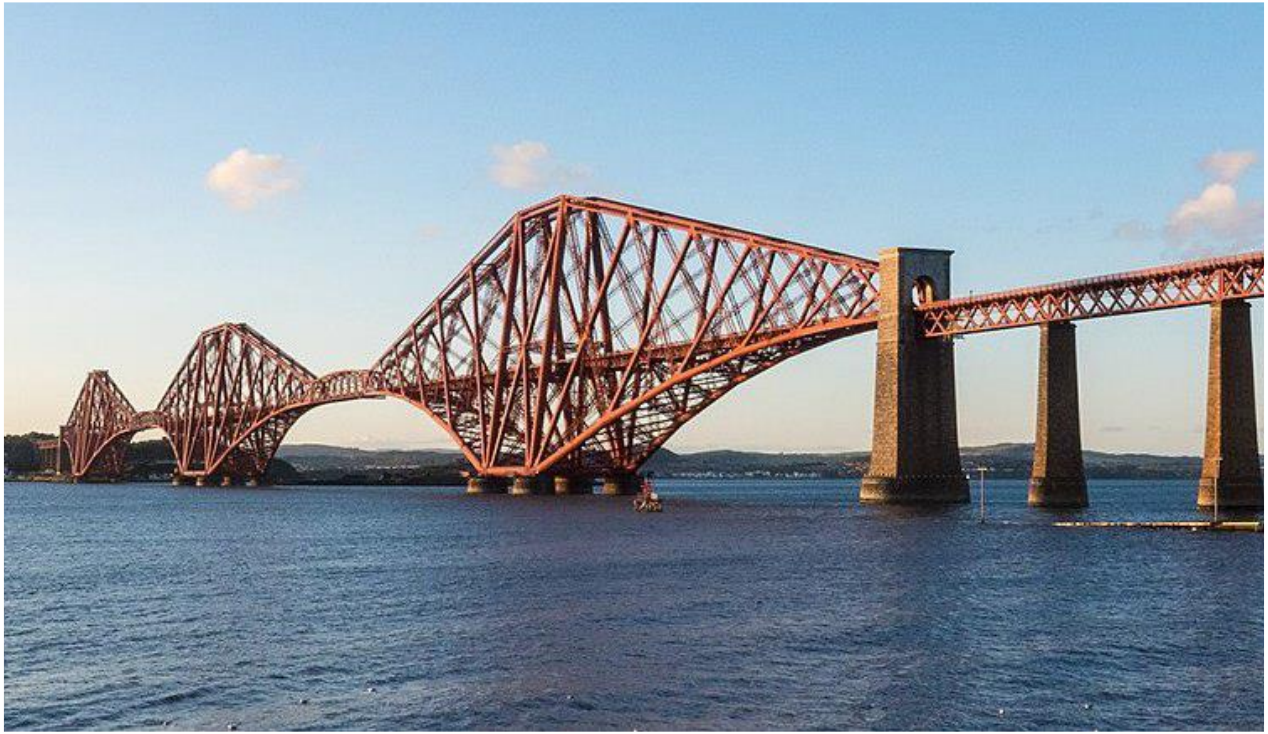


6. Trusses





6. Trusses



6. Trusses

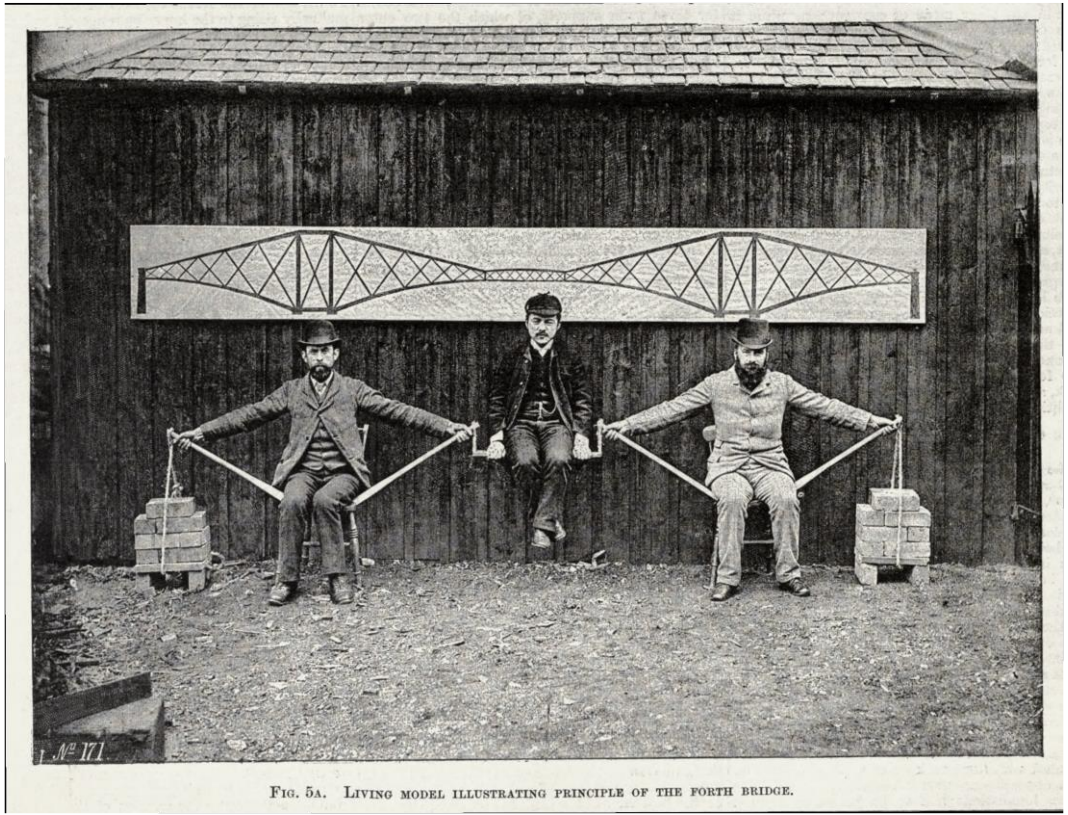


FIG. 5A. LIVING MODEL ILLUSTRATING PRINCIPLE OF THE FORTH BRIDGE.

6. Trusses

Supporting structures consisting of (straight) bars connected to each other in nodes

Ideal trusses are based on idealizing **assumptions**:

- The bar axes (geom. location of the gravity axis) are always straight
- Bar axes intersect at one point (node) each
- Pins connect bars with frictionless joints
- External forces act only on nodes

➢ Thus, only **tensile and compressive forces** act, no moment effect.

6. Trusses

Convention for bar forces:

Released bar forces are assumed to be "pointing out" of the bar:



positive bar forces S: **Tensile force**

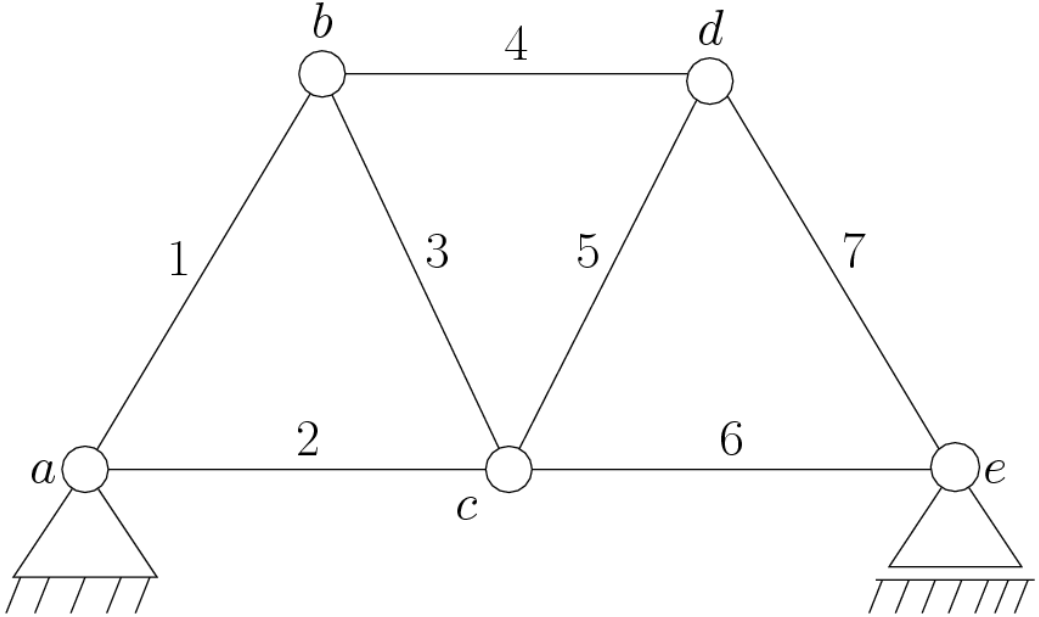
negative bar forces S: **Compressive forces**

6. Trusses

- Assumption: **plane** trusses, i.e. bars, pins and forces in one plane.
- Equilibrium conditions for cut out nodes or moment joints:
  - $\sum F_x = 0; \sum F_y = 0; (\sum M = 0$; always true, central force system)
- With **n** nodes, **b** bars and **s** support reactions, the counting criterion for plane ideal trusses is obtained:

Box 6.2

6. Trusses - Example



b = 7

s = 3

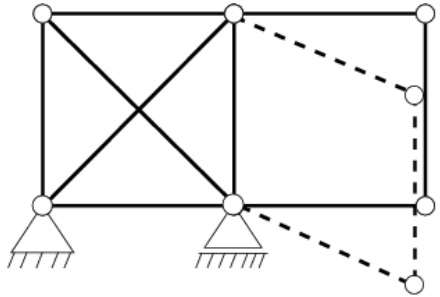
n = 5

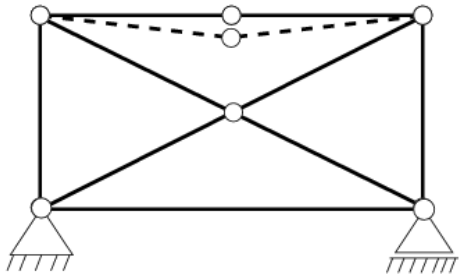
7 + 3 = 2 · 5 -> statically determined (if undisplacable)

6. Trusses

Pay attention:

- Counting criterion can be fulfilled and the truss can still be movable



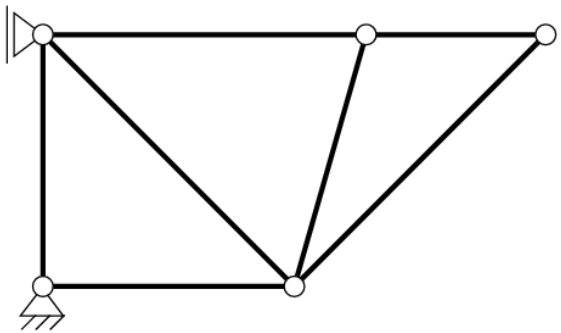


- Further necessary criterion: **kinematic determinacy**
- Kinematic determinacy by distinction into **external** determinacy and **internal** determinacy

6.1. Zero-force bar

Bars in which under certain circumstances **no force** acts are called **zero-force bars**

- can be ignored for calculation
- nevertheless, required in real construction



By cutting them free they can be detected immediately.

6.1. Zero-force bar

Typical configurations:

1. Unloaded pin with 2 non-parallel bars

Box 6.3

2. Loaded pin with 2 bars and force in the direction of one bar

Box 6.4

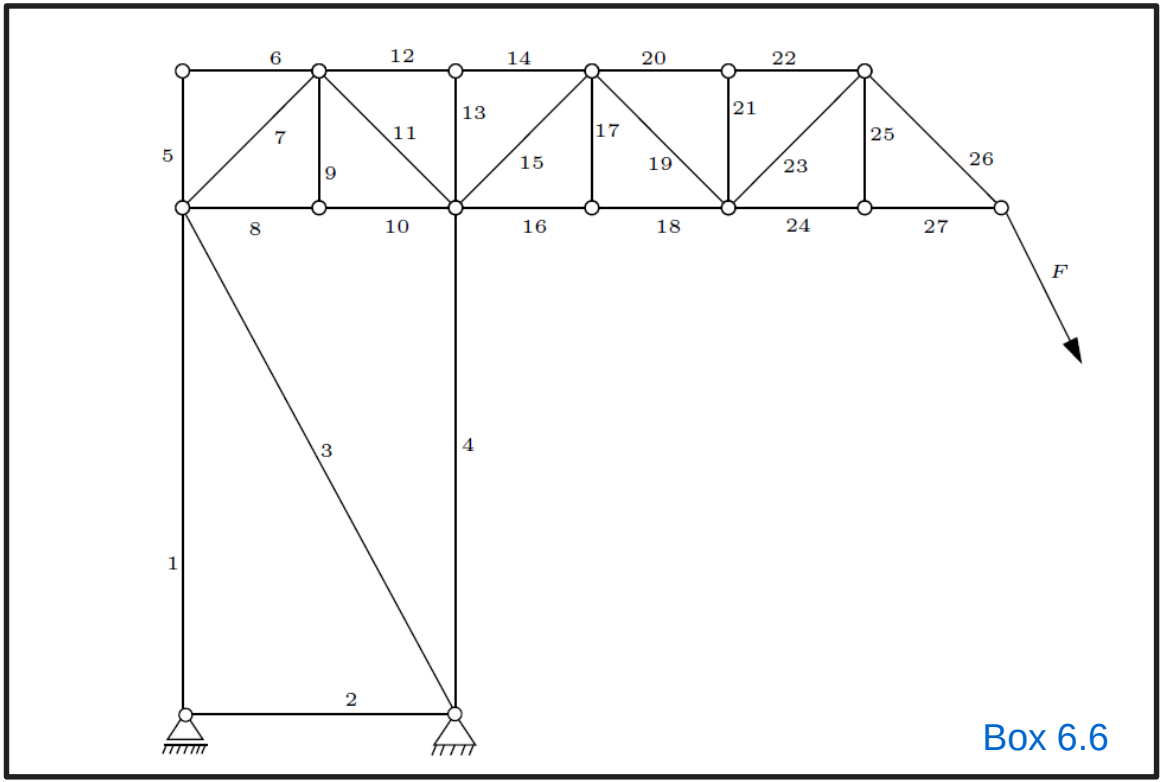
3. Unloaded pin with 3 bars, 2 of which are collinear

Box 6.5

6.1. Zero-force bar

Given.: System with load

Wanted: Zero-force bars
with criterion

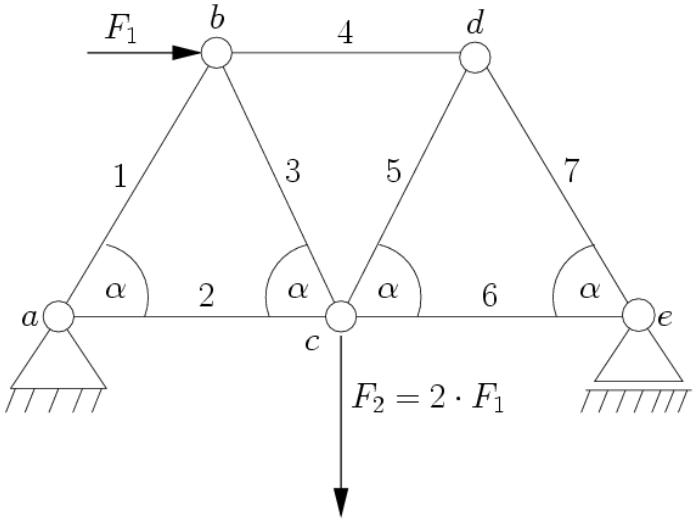


6.2.1. Computational Method of Joints

- Cutting principle at **each pin** leads to two equilibrium conditions each.
- For statically determined trusses, **all** unknown bar and support forces can be calculated with it.
- Sign convention for bars must be observed.
- For **simple** trusses (triangles only), first calculate support forces, then continue at the node with only 2 unknowns.

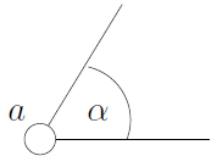
6.2.1. Computational Method of Joints

Example: **All Support forces are given**



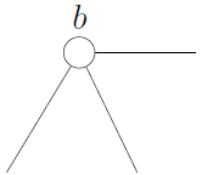
6.2.1. Computational Method of Joints

- Joint a



Box 6.7

6.2.1. Computational Method of Joints



- Joint b

Box 6.8

6.2.1. Computational Method of Joints

Pins		Bar Forces							Support Forces			Loads
		S_1	S_2	S_3	S_4	S_5	S_6	S_7	A_V	A_H	B	
a	H	cos α	1							-1		
	V	sin α							1			
b	H	-cos α		cos α	1							F1
	V	-sin α		-sin α								
c	H		-1	-cos α								
	V			sin α								-F2
d	H											
	V											
e	H											
	V											

6.2.1. Computational Method of Joints

- Computer oriented calculation : Matrix form

$$\begin{bmatrix} \cos\alpha & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ \sin\alpha & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ -\cos\alpha & 0 & \cos\alpha & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\sin\alpha & 0 & -\sin\alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -\cos\alpha & 0 & \cos\alpha & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sin\alpha & 0 & \sin\alpha & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -\cos\alpha & 0 & \cos\alpha & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\sin\alpha & 0 & -\sin\alpha & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -\cos\alpha & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \sin\alpha & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} S_1 \\ S_2 \\ S_3 \\ S_4 \\ S_5 \\ S_6 \\ S_7 \\ A_V \\ A_H \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ F_1 \\ 0 \\ 0 \\ -F_2 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Abbrivated:

$$\mathbf{K} \cdot \vec{U} + \vec{L} = \vec{0}$$